POKHARA UNIVERSITY

Level: Bachelor | Semester: Spring | Year : 2017
Programme: BE | Full Marks: 100
Course: Chemistry | Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) What is buffer solution? Define acidic and basic buffer. How do you calculate the pH value of buffer mixture? Give your answer illustrating Henderson's Equation. Calculate the pH of resulted solution when 100ml of 0.05M HCL is added to 100 ml of 0.1M ammonia?(pkb =4.74) 8

   b) Define electromechanical series with applications. What is the concentration of Ni^{2+} in the cell at 25^0c, if the emf is 0.601V? 7

$$Ni(s)|Ni^{2+}(a=?)||Cu^{2+}(0.75m)|cu(s)$$

Given

$$E^0Ni|Ni^{2+} = 0.25V \;\&E^0Cu^{2+}|Cu = 0.34V$$

2. a) Define electron affinity. Explain the factors affecting electron affinity. How electron affinity of Chlorine is higher than Fluorine. 7

   b) What are transition elements? Zinc and cadmium are not considered as transition metals give reasons. Explain the characteristics properties of transition element with regard to 8

   i. complex formation

   ii. oxidation state

3. a) Describe the structure, stability and reactions of carbocation. 8

   b) Describe S_N1 and E1 reaction with mechanism. 7

4. a) Write the preparation, properties and uses of Teflon and Polyvinyl chloride. 8

   b) What are the demerits of natural rubber? How the properties of it can be improved, explain it. 7

5. a) Give the principle of mass spectroscopy. Write the applications of TLC. 8

b) What are essential requirements of an explosive? Compare the properties of solid and liquid propellants. 7

6. a) What is green house effect? Describe the photochemistry of ozone layer depletion. 7

b) Define acid rain and alkalinity. Discuss briefly about source of water pollution, its effects and control measures. 8

7. Write short notes on: **(Any two)** 2×5

a) Electronic configuration

b) Corrotion and its mechanism and control

c) Principle of paper chromatography